Caden Cheng

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Education

University of California, San Diego — La Jolla, CA
B.S. in Computer Science and Mathematics, Data Science Minor — *Expected Grad 2028*
Relevant Coursework: Data Structures, Systems Programming, Algorithms, Operating Systems

Experience

Barobo, Inc — Davis, CA
Software Engineer Intern — *Jun. 2026 – Aug. 2026*

- Signed **CloudFront** video URLs with a configurable TTL (default **30 min**); rejects ∼**175** expired links/month
- Gated **10K** monthly video requests by account with per-user session and ownership records in **DynamoDB**
- Cut environment provisioning from 2h to 15m with **3 Terraform** modules and automated **AWS CLI** scripts
- Dropped p50 latency on the course-videos API **400→240ms** by collapsing **4** N+1 **SQL** queries into joins

Computer Science Engineering Society (CSES) at UCSD — La Jolla, CA
VP of Technology — *Sep. 2026 – Present*

- Supervise **3 engineers** shipping production **RAG** features for an enterprise AI startup client on **GCP**
- Designed the technical interview rubric and question set used to hire engineering managers and developers
- Onboarded **5** EMs on a repo-setup guide: **CODEOWNERS**, lint/build **CI**, code-owner-gated **main**
- Wrote the **Git** guide **26** developers follow for branch naming, conventional commits, and their first reviewed PR

Software Developer — *Oct. 2025 – Aug. 2026*

- Built **Google OAuth (NextAuth)** + **JWT** sessions for a lab-inventory app used by **100** members in **10** labs
- Enforced **4-role RBAC** across 12 API routes with 10 scoped permissions; 15 **Jest** tests run on every PR

Triton Web Developers (TWD) at UCSD — La Jolla, CA
Software Developer — *Mar. 2026 – Present*

- Integrated **TinaCMS** so officers edit content without a PR; **40+** member edits shipped with **0** dev commits
- Modeled the homepage as **7** typed **CMS** sections, replacing hand-edited **HTML** that broke on every edit

Projects

remark. [Link] | *Python, TypeScript, React, FastAPI, Claude API, Deepgram* | **Berkeley AI Hackathon** Jun. 2026

- Shipped in **24h** a voice agent turning overheard plans into calendar events and commands into actions
- Routed voice intents across **3 tiers** (Direct APIs → Specialized Agents → Browser Automation) covering **11 apps**
- Hit **3.2s** median speech-to-event latency (20 trials); **1.5s** of it is a deliberate silence-flush window
- Built an on-device **OpenCV** face gate (**36 ms/frame** Haar) so the mic streams only while a face is in view

inVISION [Link] | *Python, OpenCV, MediaPipe, NumPy* | **Best Interactive AI, DiamondHacks** Apr. 2026

- Cut renderer peak memory **5x** (34.7 → 6.7 MB on a 267k-face mesh, tracemalloc) via **NumPy** transforms
- Mapped **21** MediaPipe hand landmarks to a smoothed rotation matrix: **23 ms/frame** detect, ∼**64 ms** max lag
- Profiled post-hackathon: per-face np.cross loop, not the transform, costs **664 ms/frame** on 10k faces
- Ran a **YOLOv8** detect → crop → background-removal pass on uploads before **TripoSR** mesh generation

Silent Speech [Link] | *PyTorch, MediaPipe, OpenCV* | **#1 Overall Best Hack, SanD Hacks** Jan. 2026

- Trained a **BiGRU** on 90-frame clips of **88** mouth/jaw landmarks (of 478); **78%** (39/50) live word accuracy
- Noise + frame-drop augmentation raised val acc **72→79%** (3-seed avg); mixup dropped after no gain
- Recorded and labeled the 5-word landmark dataset in **24h**; normalized by mouth width for distance invariance
- Cut feature extraction **5x** (0.25 → 0.05 ms/frame) by dropping the 478-point mesh to the **88-point** subset

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, C#, HTML/CSS, SQL
Frameworks & Domains: React, Next.js, Node.js, Tailwind CSS, PyTorch, TensorFlow, LLMs/GenAI, NLP, RAG
Developer Tools: Git/GitHub, Docker, CI/CD, AWS (CloudFront, DynamoDB, CLI), GCP, Terraform, Cursor, Linux/Unix
Libraries: Pandas, NumPy, Matplotlib, Torchvision, OpenCV, MediaPipe, Mongoose